

Diffuseness and Weak-Interior Obstructions for Strong Regularity with the r-BWOP

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Source Question

Lopez-Perez, Martinez Vano, and Rueda Zoca introduce the r-big weak open property and ask in Remark 3.3(3):

“A natural question, whose answer we do not know, is whether there are Banach spaces with the r-BWOP and being strongly regular.”

The question is on PDF page 14 of arXiv:2510.15461 and at line 671 of the parsed source file `Big_weak_open_radius_versus_big_slice_diameter.tex` [1].

14

G. LÓPEZ-PÉREZ, E. MARTÍNEZ VAÑÓ, A. RUEDA ZOCA

there exists a sequence of Banach spaces X_n isomorphic to X such that $\ell_1(\mathbb{N}, X_n)$ satisfies the conclusion of the theorem. This reveals a wide class of non-isomorphic Banach spaces satisfying the conclusion of the theorem. This should be compared with the examples constructed in [18, Theorem 3.1] of Banach spaces with the r-BSP and such that the infimum of the diameter of slices is 1, which are based on very specific tree spaces.

- (2) At this point it is natural to wonder whether the above mentioned spaces constructed in [18, Theorem 3.1] could have been used to distinguish the r-BWOP and the D2P. In fact, the answer is no, because all the spaces constructed there satisfy the CPCP, in particular, the unit ball of all these spaces contains non-empty relatively weakly open subsets with arbitrarily small radius.
- (3) As we have pointed out before, in [18] Banach spaces with the r-BSP and the CPCP are constructed. A natural question, whose answer we do not know, is whether there are Banach spaces with the r-BWOP and being strongly regular.

ACKNOWLEDGEMENTS

Definitions

For a bounded set A in a Banach space X , write

$$r(A) = \inf\{t > 0 : A \subseteq x + tB_X \text{ for some } x \in X\}.$$

The space X has the *r-big weak open property* (r-BWOP) when every nonempty relatively weakly open subset of B_X has radius one [1, Definition 2.3]. Recall that

$$r(A) \leq \text{diam}(A).$$

The space X is *strongly regular* when every nonempty bounded convex subset admits finite convex combinations of slices of arbitrarily small diameter. We use the standard localization of this definition: if X is strongly regular, W is a nonempty relatively weakly open subset of B_X , and $\varepsilon > 0$, then there are nonempty relatively weakly open subsets $U_1, \dots, U_n$ of B_X , all contained in W , and positive weights λ_i summing to one such that

$$\text{diam} \left(\sum_{i=1}^n \lambda_i U_i \right) < \varepsilon.$$

This localization is recorded, in the more general topological formulation, in [2, Lemma 2.2].

Finally, X has property $(\overline{\text{PI}})$ when, for every finite convex combination C of slices of B_X and every $x \in C$, there is a relatively weakly open neighborhood V of x in B_X such that $V \subseteq \overline{C}$, where the bar denotes norm closure [3, Definition 2.1].

Proof Intuition

Each weak opening U_i has radius one and therefore diameter at least one. If all other coordinates in a Minkowski convex combination are frozen, the combination contains a translated copy of $\lambda_i U_i$. Thus a combination of diameter d cannot assign any constituent weight larger than d . Arbitrarily small strong-regularity witnesses must consequently spread their mass over arbitrarily many weak openings. Moreover, a combination of diameter below one cannot contain a weak opening even after norm closure, because that opening would have radius below one. Property $(\overline{\text{PI}})$ forces exactly such an opening for every convex combination of slices, so it is incompatible with strong regularity and r-BWOP together. The original unrestricted existence question remains open.

Main Partial Result

Theorem 1 (Diffuseness and weak-interior obstruction). *Let X have the r-BWOP. Let $U_1, \dots, U_n$ be nonempty relatively weakly open subsets of B_X , let $\lambda_1, \dots, \lambda_n > 0$ satisfy $\sum_i \lambda_i = 1$, and put*

$$C = \sum_{i=1}^n \lambda_i U_i.$$

Then:

- (i) $\text{diam}(C) \geq \max_i \lambda_i$;
- (ii) if $\text{diam}(C) < \varepsilon$, then $\lambda_i < \varepsilon$ for every i and $n > 1/\varepsilon$;
- (iii) if $\text{diam}(C) < 1$, then $\overline{C}$ has empty interior in the relative weak topology of B_X .

If, in addition, X is strongly regular, then for every nonempty relatively weakly open $W \subseteq B_X$ and every $\varepsilon \in (0, 1)$ there is such a combination with $U_i \subseteq W$ and diameter below ε ; every one of its coefficients is below ε , it uses more than $1/\varepsilon$ terms, and its norm closure has empty relative weak interior.

Proof. Fix i . For every $j \neq i$, choose a point $u_j \in U_j$. If $u, v \in U_i$, then both

$$\lambda_i u + \sum_{j \neq i} \lambda_j u_j \quad \text{and} \quad \lambda_i v + \sum_{j \neq i} \lambda_j u_j$$

belong to C , and their distance is $\lambda_i \|u - v\|$. Hence

$$\text{diam}(C) \geq \lambda_i \text{diam}(U_i) \geq \lambda_i r(U_i) = \lambda_i,$$

where the last equality is the r -BWOP. Taking the maximum over i proves (i).

If $\text{diam}(C) < \varepsilon$, part (i) gives $\lambda_i < \varepsilon$ for every i . Therefore

$$1 = \sum_{i=1}^n \lambda_i < n\varepsilon,$$

which proves (ii).

For (iii), suppose that a nonempty relatively weakly open subset V of B_X were contained in $\overline{C}$. Then

$$1 = r(V) \leq \text{diam}(V) \leq \text{diam}(\overline{C}) = \text{diam}(C) < 1,$$

a contradiction.

Now assume that X is strongly regular. Apply the standard localization statement above inside W to obtain nonempty relatively weakly open $U_i \subseteq W$ whose convex combination has diameter below ε . The remaining assertions follow from (ii) and (iii). $\square$

Theorem 2 (A broad no-go class). *No Banach space can simultaneously be strongly regular, have the r -BWOP, and have property $(\overline{\text{P1}})$.*

Proof. Suppose that X has all three properties. Strong regularity applied to B_X gives a finite convex combination C of slices of B_X with $\text{diam}(C) < 1$. Choose $x \in C$. Property $(\overline{\text{P1}})$ gives a nonempty relatively weakly open neighborhood V of x in B_X with $V \subseteq \overline{C}$. Therefore

$$1 = r(V) \leq \text{diam}(V) \leq \text{diam}(\overline{C}) = \text{diam}(C) < 1,$$

which is impossible. $\square$

Corollary 3. *No isometric predual of $\ell_1(\Gamma)$ can be both strongly regular and have the r -BWOP.*

Proof. Every isometric predual of an ℓ_1 space has property $(\overline{\text{P1}})$ [3, Corollary 2.8]. Apply Theorem 2. $\square$

Verification Notes

The core estimate is an exact scaled-copy argument; it does not use compactness, separability, attainment of the diameter, or scalar restrictions. The closure in Theorems 1 and 2 is norm closure. Since C is convex, its norm closure is also its weak closure, but that identification is not needed: only equality of diameters under norm closure is used.

No computation is required. The two imported facts to check are the standard localization of strong regularity and the theorem that isometric ℓ_1 -preduals have $(\overline{\text{P1}})$; both are cited at their source locations.

Limitations and Novelty Check

This is a partial result only. It neither supplies a strongly regular r-BWOP space nor rules one out without an additional local-interior property. The direct full contradiction stops precisely because arbitrary convex combinations of slices need not contain relatively weakly open subsets, even up to closure.

The run registry and solution indexes were searched for arXiv:2510.15461 and the phrases “r-BWOP strongly regular,” “r-big weak open property strongly regular,” and close variants. A bounded external search of arXiv and the June 2026 version-of-record page was also performed through 9 August 2026. No later answer to the exact question was found, and the version of record still states it as open. The $(\overline{\text{P1}})$ obstruction is a direct new synthesis of the source question with the known local-interior property; novelty confidence is moderate, and the packet makes no claim of priority.

Human Review Recommendation

Review as a likely-valid structural partial. The most useful checks are:

1. the passage from a constituent U_i to the translated scaled copy $\lambda_i U_i \subseteq C$;
2. applicability of the classical localization lemma to the stated definition of strong regularity;
3. the convention that $\overline{C}$ in $(\overline{\text{P1}})$ is norm closure; and
4. the cited $(\overline{\text{P1}})$ result for isometric ℓ_1 -preduals.

References

- [1] G. Lopez-Perez, E. Martinez Vano, and A. Rueda Zoca, *Big weak open radius versus big slice diameter*, arXiv:2510.15461; Mediterr. J. Math. **23** (2026), article 141, <https://doi.org/10.1007/s00009-026-03143-x>.
- [2] G. Lopez-Perez and R. Medina, *The equivalence between CPCP and strong regularity under Krein–Milman property*, arXiv:2305.18976.
- [3] J. Becerra Guerrero and F. J. Fernandez-Polo, *Relatively weakly open convex combinations of slices and scattered C^* -algebras*, arXiv:1902.09209.